

Inspiring Excellence

BRAC University

Department of Computer Science and Engineering

CVIS Hands-on Training Project

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Textual Emotion Recognition

Enabling Machines to Understand Human Emotions using LSTM, CNN, and Hybrid CNN-LSTM

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1. Project Overview

When people read a sentence like *"I'm so happy today!"*, they immediately recognize the emotion behind it. But can a computer understand this? That's what I wanted to find out.

In this project, I tried to create a system that can detect emotions from text. The system can identify six emotions: **joy, sadness, anger, fear, love, and surprise**. My goal is to teach computers to understand the emotions that people express in their writing.

2. Dataset: Collection and Composition

The dataset used in this project was collected from Kaggle, it's called the *Emotion Dataset* (https://www.kaggle.com/datasets/parulpandey/emotion-dataset). It has 20,000 text samples where each one is labeled with an emotion.

The data was divided into three parts:

- **Training Set (16,000 samples):** Used to teach the model.
- **Validation Set (3,000 samples):** Used to adjust the model's settings.
- **Test Set (1,000 samples):** Used to check how well the model performs.

3. Data Preprocessing

Raw text data often contains noise such as extra punctuation, inconsistent capitalization, and unnecessary symbols. Therefore, several preprocessing steps were applied before putting it to my models. Here's what I did:

1. **Lowercasing:** I made everything lowercase so "HAPPY" and "happy" look the same to the computer.
2. **Special Character Removal:** I took out things like "!?@#" that don't help much.
3. **Stemming:** I reduced words to their base form. For example, "running" becomes "run".
4. **Tokenization:** Since computers don't understand words directly, I converted text into numbers. Each word gets a number, and i make sure all sentences are the same length by adding padding.

4. Model Architectures

Three types of deep learning models were developed and compared to identify the most effective approach:

Bi-directional LSTM (LSTM Model)

LSTM stands for Long Short-Term Memory. It is capable of remembering previous information while reading text. This model reads text word by word and keeps track of what came before. The "bi-directional" part means it reads forward AND backward, so it gets the full picture. It's like reading a sentence and then reading it again backwards to catch anything which are missed.

1D CNN (Convolutional Neural Network)

CNNs are famous for recognizing images, but they can work with text too! In this project, the CNN model was used to detect important word patterns that represent specific emotions, such as "so angry" or "really happy."

Hybrid CNN-LSTM Model

This model combines CNN and LSTM. The CNN part finds important word patterns, and then the LSTM part understands how those patterns connect together.

Ensemble Methods:

To further enhance model performance, multiple models were combined using ensemble techniques. Five ensemble methods were tested:

- **Hard Voting:** Each model casts a vote, and the majority prediction is selected.
- **Soft Voting:** Probabilities from all models are averaged to determine the final prediction.
- **Weighted Voting:** Models with higher accuracy are given greater influence.
- **Stacking:** A meta-model (Logistic Regression) learns from the predictions of the base models.
- **Max Confidence:** The prediction with the highest confidence score is chosen.

5. Results:

The models achieved the following accuracies on the test set:

Model	**Accuracy**
LSTM Model	88.80%
CNN Model	**89.80%**
Hybrid CNN-LSTM	87.80%
Soft Voting Ensemble	**90.70%**
Weighted Ensemble	90.60%
Stacking (Logistic Regression)	88.90%

Observation: Among individual models, the CNN performed best with an accuracy of 89.80%. However, the *Soft Voting Ensemble* achieved the highest overall accuracy of 90.70%, demonstrating the benefit of combining models.

6. Future Improvements

The performance of the system could be improved by:

- Using pre-trained word embeddings such as GloVe or Word2Vec.
- Fine-tuning transformer-based models like BERT or DistilBERT.
- Incorporating explainability tools to interpret model predictions.

7. Conclusion

This project presented a text-based emotion recognition system that achieved an accuracy of 90.70% using an ensemble of deep learning models. The system can be applied in various fields, including:

- **Social Media Monitoring:** Understanding public sentiment.
- **Customer Support:** Detecting user frustration in service interactions.
- **Mental Health:** Identifying emotional distress in written communication.
- **Content Moderation:** Recognizing aggressive or hateful language.

The results highlight the potential of deep learning methods in helping computers analyze and interpret human emotions through text.